EVIDENCE COMPANION / 01 · CLAIMS AND BOUNDARIES

An inspectable chain from purpose to reuse.

AGI ALPHA FLYWHEEL — When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

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Principal achievement: observed executable transfer. A frozen quantitative instrument recovered through the existing research lifecycle executes correctly on fresh, explicitly scoped microscopy missions. Its accepted quality matches competent established analysis. Commercial viability and retained economic advantage remain conditional.

Claim-to-evidence map

Claim and type	Finding and limit	Inspect from the study root
Instrument construction *Observed*	Source-bound native moments and explicit affine geometric pushforward; established mathematics.	`src/microscopy_transfer.py` `protocols/source-freeze.json`
Fresh recipient execution *Observed*	48/48 compatible missions; 12/12 incompatible requests refused. Two partitions in one fresh image/timepoint.	`confirmation/results/frozen-v1/summary.json` `confirmation/RESULTS.md`
Competent alternatives *Observed*	Conventional analysis: 48/48. Identical ordinary-library bytes: 48/48 exact scientific parity. No graph-specific gain.	`run_confirmation.py` `confirmation_reference.py`
Same-model comparisons *Observed*	U, E and E+ each 12/12. One worker per arm; same tools and ceilings, incomplete cost telemetry.	`hosted_trials/scoring.json` `hosted_trials/pretrial_commitment.json`
Lifecycle and accounting *Executed fixtures*	32/32 controls using 29 unchanged upstream files. Production microscopy scope is refused. No live ENS.	`integration/results/frozen-v1/summary.json` `integration/source_manifest.json`
Economic viability *Simulated*	13,824 histories; cent-exact cash identities; eight accounting tests. Inputs remain hypotheses.	`economics/results/results.json` `economics/parameter_registry.json`
Actual operating result *Observed / unknown*	Zero customers, collections and paid API calls. Complete actual research cost and profit are unknown.	`integration/results/frozen-v1/native_actual_unknown.json`

Reading the evidence correctly

"Accepted" refers to the declared numerical or semantic test unless explicitly described otherwise. It is not customer sign-off. The common scorer is outside the three hosted workers but inside the same AI-assisted team; it is not independent external validation. The 48 missions are authored maps over two partitions, not 48 independent biological specimens.

The instrument measures supplied label support. A label can contain disconnected components and can touch a source boundary. It does not validate biological cell counts, segmentation quality, physical registration, nonlinear maps, intensity measurements or clinical meaning. Numerical error bounds are software agreement, not microscope precision.

Evidence vocabulary

Observed: executed outputs, bytes, timings and refusals. **Sourced**: official specifications, public data provenance and a human-service rate card. **Assumed/calibrated**: workload, labor, buyer values, probabilities and service scope. **Simulated**: every customer decision, payment and commercial outcome. A simulation interval describes variation within those assumptions.

EVIDENCE COMPANION / 02 · PURPOSE, DATA AND PROTOCOL

Freeze the instrument. Recheck the mission.

The transformation witness

Original purpose. Preserve supplied label identities, native physical measure and centroids, and report an explicitly requested affine geometric pushforward.

Binding restriction under examination. Require all quantitative measurement to pass through a common target raster.

Why removable. Affine maps act directly on voxel moments; the useful quantities do not require re-rasterization. This is an established analytical alternative, not an invention of new affine mathematics.

Replacement and realization. Own one bounded NPY byte snapshot, bind its SHA-256 to the contract, validate units/axes/meaning, summarize native support and transform its moments.

Falsifier. A valid, supported mission fails fixed numerical tolerances; changed or unsupported meaning is accepted; or the claimed advantage disappears against an equally competent baseline. The last falsifier applies to superiority: conventional analysis matches accepted quality.

Public data and rights

	Development	Fresh confirmation
IDR study / image	IDR0062A / 6001240	IDR0052A / 5514375
Partition	Published labels, 61 regions	Cell and Chromosomes; each label 8
Selection	Native labels, channel 0	Time index 7, channel 0, fixed in advance
Shape, z/y/x	$236 \times 275 \times 271$	$31 \times 256 \times 256$ per group
Spacing, μm	0.5002025532 / 0.3603981535 / 0.3603981535	0.75 / 0.2516 / 0.2516
Credit and dataset	Blin et al.; 10.17867/10000125	Walther et al.; 10.17867/10000123

Both sources are CC BY 4.0. Original compressed chunks and metadata are retained; exported NumPy arrays remove declared singleton/time axes without altering labels. Offline reconstruction reproduces all three export hashes. The archive-inventory addendum reconciles a study-page snapshot discrepancy without replacing the original receipt.

Prospective sequence and acceptance

Held task bytes were committed on 27 October 2026 before the 17:40:14 UTC source freeze; first evaluation completed at 17:42:57 UTC. Fresh sources did not revise the frozen instrument. Each recipient invocation uses a new Python process. The conventional comparator may cache native summaries; the ordinary directory receives identical artifact bytes and permissions.

Acceptance requires exact label IDs and voxel counts, native/mapped measure relative error at most 10^{-9}, and centroid error at most 10^{-8} μm. The separate reference transforms every occupied voxel center and uses a scalar triple product for mapped volume. Maximum recovered discrepancies are 4.3964×10^{-16} relative measure and 1.0737×10^{-11} μm centroid.

Identity and source commitments

Frozen core SHA-256:

`a04326ce0504bfe3ac290b6cb65ee6f089cc66a607f70b7cc7869099250c77db`

Held-task SHA-256:

`a340f72f6efe4ec41f0d0b51de14f9131989318f6fe89818c680fcf46509fbd5`

Upstream main commit:

`47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4`

Complete data hashes: `data/development/idr0062/manifest.json` and `data/confirmation/idr0052/manifest.json`.

EVIDENCE COMPANION / 03 · ECONOMICS AND ADMISSION

Profit is a scenario. Retention is a decision.

Sources, assumptions and what is actually priced

The opened CHUM rate card lists formatted image analysis at $143.50/hour for businesses and $129.75/hour for external university/hospital customers. CAD is inferred from the Montreal provider; the page's dollar sign is not a project quotation. Assumed equivalent-service hours are 1.5, 2.5 or 4. The human alternative remains the primary commercial reference.

A frozen development search uses 24 worlds, prices CAD200/300/425, deposits 0/50%, participant thresholds 1/3 and stopping. It selects the established route, CAD425, no deposit. Confirmation uses 96 fresh worlds per route/arm/scenario: 13,824 histories. Primary retained benefit is zero; no model upgrade, prompting gain or shared-graph advantage is silently inserted.

Primary established-route ledger, mean CAD

Quantity	Amount	Meaning
Net collections	1,160.94	Simulated accepted balances collected
A: paid-job delivery cash	534.14	Includes origin, failures, correction and support
B / C for E	0.00 / 0.00	No additional retention or practice
Settled cash contribution	626.80	Collections less cash expenses
Owner opportunity cost	29.94	Separate from cash expense
Economic surplus	596.86	Pretax modeled contribution less opportunity cost
Of which fictional origin	220.00	CAD550 collected less CAD310 A and CAD20 opportunity cost
Of which subsequent work	376.86	Fresh-work modeled economic contribution
Ending available funds	1,976.80	Includes CAD1,500 seed; excludes reserve/obligations

Four of 96 primary worlds lose economic value even with the positive supplied origin. The mean interval [520.92, 672.80] is Monte Carlo variation under assumptions. It is not an estimate of actual business profitability. No actual sale funds this research.

Selective retention: calculated thresholds, assumed benefits

Scenario	E+ minus E	Decision or implication
Zero retained saving	−5.00	Stop after screening; no extra Alpha
Assumed 30 minutes saved/use	−21.17	Forecast reuse exceeds realized demand
Assumed 60 minutes saved/use	+32.66	Conditional positive value; saving unmeasured
False admission; 30 minutes added/use	−128.83	Negative transfer and costs retained

With B = CAD5 assessment + CAD55 retention + CAD15 upkeep, break-even avoidable cash saving is $75/(65N)$ hours/use. It is 17.31 minutes at four *realized eligible* reuses, 34.62 at two. An identical ordinary library matches E+ in every history. Optional Gym is separate C; it is rejected in the primary case.

Cash and capacity qualifications

Refundable deposits remain liabilities; delayed invoices are not available cash. Default, partial delivery, withdrawal, common failure, refunds and support are modeled separately. Low opening capital can stop a profitable-looking project. The buyer rule conservatively subtracts the full quote, not exact expected payment. The 24-hour capacity bound covers fresh delivery/correction/support only; fixed construction, coordination, retention and practice require separately provisioned capacity and unmodeled lead time.

Inspect: `economics/parameter_registry.json` (57 labeled inputs), `economics/results/example_ledger.json`, `economics/results/confirmation.csv`, `economics/test_accounting.py`. Source: CHUM rates and service scope.

EVIDENCE COMPANION / 04 · AUDIT, REUSE AND NEXT TEST

Preserve what worked. Keep the failed hypotheses.

Correction and negative-result history

Stage	Preserved result and response
Before source freeze	Source binding/mutable-mmap risk was found in review; one owned byte snapshot replaced it. The old race was not experimentally reproduced. NPZ, empty-file and payload checks were hardened; signed nonzero labels were restored to scope.
Development execution	A NumPy Boolean blocked diagnostic JSON serialization; numerical outputs were retained and serialization fixed. Installed Black failed to import a dependency; no Black pass is claimed.
Named raster diagnostic	At target spacing 0.75 / 1.5 / 3 μm, 6 / 26 / 51 of 61 regions exceed 1% volume error. This specific raster pipeline is not the U arm or a limit on frontier intelligence.
Strong analytical baseline	Matches all accepted fresh quantities; cached preparation plus mapping totals about 0.333 seconds. Recipient external invocations total 11.542 seconds. Timers cover different deployment overheads; no speed advantage is claimed.
Post-run evaluator audit	Nonfinite/duplicate-ID checks were strengthened because a Python maximum could mask NaN. Stored outputs were finite; strict rescoring leaves acceptance unchanged. Four NaN probes verify the guard. Frozen core and first outputs remain intact.
Commercial ablation	Mean E surplus: established 596.86, technical-only 356.86, coordination-only 566.24, combined 356.24 CAD. The supplied joint arrangement does not create a superior market here.

Reuse without inherited authority

The integration exercises unchanged Journal, TrustRegistry and CostLedger components plus the existing research lifecycle. Actual signatures and authenticated encryption are tested with labeled fixture identities. Unknown members, revoked rights, changed bytes, altered mission meaning and duplicate A/B/C activity charges are refused. The native protected-capsule interface refuses this unsupported scientific scope. Production admission has not been obtained.

A successor inherits useful executable structure. It must still check its source, dependencies, input meaning, requested outputs and acceptance. Authority is a separate grant. Revocation prevents new mediated access; it cannot erase copies already retrieved. Rights-cleared data and code can be shared; secrets and protected evaluation material cannot.

Smallest next real-world experiment

Arrange four eligible unfamiliar quantitative-report briefs, with an independent domain reviewer and a scope-matched human-service quotation. Compare E and E+ using the same accepted quality, deadlines, tools and customer obligations; randomize order where feasible. Record source validation, adaptation, verification, correction, customer effort, support and collections. Distinguish billable cash avoided from idle salaried time. Advance retention only if measured reusable benefit exceeds complete B costs; revise or stop if ordinary expert tools remain equivalent.

Foundations, software and disclosure

The frozen source papers are *Transformational Intelligence: Learning to Change the Game*; *Invent the Economy: Transformational Intelligence for Creating Viable Markets*; and *Transformational Autonomous Enterprises: SUCCESSOR Ω Harness*. Their synthetic result boundaries and cost distinctions are preserved in `integration/foundation_audit.md`. They supply design mechanisms, not validation of this market.

Recorded software: Python 3.12.14, NumPy 2.3.2, SciPy 1.17.0, Numcodecs 0.16.3, Node 24.22.1; the repository asks for Node 22.17.1. These are local research checks, not full release qualification. AI agents assisted source research, coding, experiments, review and writing. No private medical records, proprietary model-weight training or live customer contract was used. All counterparty behavior in the economics is simulated.

Primary references: OME-NGFF coordinate conventions; SimpleITK transforms and resampling; ITK spatial objects; Blin et al., 10.1371/journal.pbio.3000388; Walther et al., 10.1083/jcb.201801048. Complete source URLs, hashes and evidence are in the executable edition.